

deep learning 5

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1 Bias

The bias of an estimator $\hat{\theta}_m$ is defined as $\text{bias}(\hat{\theta}_m) = \mathbb{E}[\hat{\theta}_m] - \theta$. An estimator $\hat{\theta}_m$ is unbiased if $\text{bias}(\hat{\theta}_m) = 0$ and asymptotically unbiased if $\lim_{m \rightarrow \infty} \text{bias}(\hat{\theta}_m) = 0$.

Suppose $\{x^{(j)}\}_{j \in [m]}$ are iid generated by Bernoulli with parameter θ . Take $\hat{\theta}_m = \frac{1}{m} \sum_{i=1}^m x^{(i)}$ to be the mean. We have $\text{bias}(\hat{\theta}_m) = 0$ by Linearity of Expectation. Sample variance of Gaussian:

$$\begin{aligned}\mathbb{E}[\hat{\sigma}_m^2] &= \frac{1}{m} \sum_{j=1}^m \mathbb{E}[(x^{(j)} - \hat{\mu}_m)^2] = \frac{1}{m} \sum_{j=1}^m \mathbb{E}[(x^{(j)})^2] - 2\mathbb{E}[x^{(j)}\hat{\mu}_m] + \mathbb{E}[\hat{\mu}_m^2] \\ &= \mathbb{E}[x^2] - \frac{2}{m} \mathbb{E} \left[x^{(1)} \left(x^{(1)} + \sum_{j=2}^m x^{(j)} \right) \right] + \frac{1}{m^2} \mathbb{E} \left[\left(\sum_{j=1}^m x^{(j)} \right)^2 \right] \\ &= \mathbb{E}[x^2] - \frac{2}{m} \mathbb{E}[x^2] - \frac{2}{m} (m-1) \mathbb{E}[x]^2 + \frac{1}{m^2} (m \mathbb{E}[x^2] + (m^2 - m) \mathbb{E}[x]^2) \\ &= (\sigma^2 + \mu^2) - \frac{2}{m} (\sigma^2 + \mu^2) - \frac{2}{m} (m-1) \mu^2 + \frac{1}{m^2} (m(\sigma^2 + \mu^2) + (m^2 - m) \mu^2) \\ &= \frac{m-1}{m} (\sigma^2 + \mu^2) - \frac{m-1}{m} \mu^2 = \frac{m-1}{m} \sigma^2.\end{aligned}$$

Thus, $\text{bias}(\hat{\sigma}_m^2) = -\frac{1}{m} \sigma^2$. Then the unbiased sampler is $\tilde{\sigma}_m^2 = \frac{1}{m-1} \sum_{j=1}^m (x^{(j)} - \hat{\mu}_m)^2$.

Variance of mean estimator of Bernoulli.

$$\text{Var}(\hat{\theta}_m) = \text{Var} \left(\frac{1}{m} \sum_{j=1}^m x^{(j)} \right) = \frac{1}{m^2} \cdot m \text{Var}(x) = \frac{1}{m} \theta(1 - \theta).$$

MSE = Var + Bias²:

$$\begin{aligned}\text{bias}(\hat{\theta}_m)^2 + \text{Var}(\hat{\theta}_m) &= (\mathbb{E}[\hat{\theta}_m] - \theta)^2 + \mathbb{E}[(\hat{\theta}_m)^2] - \mathbb{E}[\hat{\theta}_m]^2 \\ &= \mathbb{E}[\hat{\theta}_m^2] - 2\theta \mathbb{E}[\hat{\theta}_m] + \theta^2 + \mathbb{E}[(\hat{\theta}_m)^2] - \mathbb{E}[\hat{\theta}_m]^2 = \mathbb{E}[(\hat{\theta}_m - \theta)^2]\end{aligned}$$

Weak Law of Large Numbers. Let $X_1, X_2, \dots$ be iid rv with $EX = \mu < \infty$. Then for all $\epsilon > 0$, $\lim_{n \rightarrow \infty} P(|\bar{X} - \mu| \geq \epsilon) = 0$.

Proof: Assume $\text{Var}(X) < \infty$. Then

$$P(|\bar{X} - \mu| \geq \epsilon) \leq \frac{\text{Var}(\bar{X})}{\epsilon^2} = \frac{\text{Var}(X)}{n\epsilon^2}.$$

Claim: Let $X_n, X \leq \mathbb{Z}_{\geq 0}$ be rvs. Then $X_n \xrightarrow{d} X$ iff $\lim_{n \rightarrow \infty} P_{X_n}(k) = P_X(k)$ for all $k \in \mathbb{Z}_{\geq 0}$.

Proof: Suppose $X_n \xrightarrow{d} X$, so for all $k \in \mathbb{Z}_{\geq 0}$, we have $\lim_{n \rightarrow \infty} F_{X_n}(k) = F_X(k)$. Write $F_{X_n}(k) = \sum_{j=0}^k P_{X_n}(j)$, so $\lim_{n \rightarrow \infty} \sum_{j=0}^k P_{X_n}(j) = \sum_{j=0}^k P_X(j)$. For $k = 0$, we have $\lim_{n \rightarrow \infty} P_{X_n}(0) = P_X(0)$. Suppose the result for $k - 1$. Then write $\lim_{n \rightarrow \infty} \sum_{j=0}^k P_{X_n}(j) = \lim_{n \rightarrow \infty} \sum_{j=0}^{k-1} P_{X_n}(j) + \lim_{n \rightarrow \infty} P_{X_n}(k)$. By induction, we have $\lim_{n \rightarrow \infty} P_{X_n}(k) = P_X(k)$.

Conversely, if $\lim_{n \rightarrow \infty} P_{X_n}(k) = P_X(k)$ for all k , then

$$\begin{aligned} \lim_{n \rightarrow \infty} F_{X_n}(k) &= \lim_{n \rightarrow \infty} \sum_{j=0}^k P_{X_n}(j) = \sum_{j=0}^k \lim_{n \rightarrow \infty} P_{X_n}(j) \\ &= \sum_{j=0}^k P_X(j) = F_X(k). \end{aligned}$$

Claim: Let X be a rv and Y_n be rvs such that $\mathbb{E}Y_n = 1/n$ and $\text{Var}(Y_n) = \sigma^2/n$ for some constant $\sigma > 0$. Define $X_n = X + Y_n$. Then $X_n \xrightarrow{p} X$.

Proof: We have

$$\begin{aligned} P(|X_n - X| \geq \epsilon) &= P(Y_n \geq \epsilon) + P(Y_n \leq -\epsilon) = P(Y_n - 1/n \geq \epsilon - 1/n) + P(Y_n - 1/n \leq -\epsilon - 1/n) \\ &\leq P(Y_n - 1/n \geq \epsilon - 1/n) + P(Y_n - 1/n \leq -\epsilon + 1/n) \\ &= P(|Y_n - 1/n| \geq \epsilon - 1/n) \leq \frac{1}{(\epsilon - 1/n)^2} \text{Var}(Y_n - 1/n) \\ &= \frac{\sigma^2}{n(\epsilon - 1/n)^2}. \end{aligned}$$

Pick $n \geq 2/\epsilon$. Then

$$P(|X_n - X| \geq \epsilon) \leq \frac{4\sigma^2}{n\epsilon^2} \rightarrow 0,$$

since σ, ϵ are constant.

Claim: Let rvs Z_n, Z be such that $Z_n \xrightarrow{p} Z$. Then $Z_n \xrightarrow{d} Z$ as well.

Proof: Suppose $F_Z(z)$ is continuous at z_0 . Let $\epsilon > 0$. Pick $\delta > 0$ such that $|F_Z(z) - F_Z(z_0)| \leq \epsilon$ for $|z - z_0| \leq 2\delta$. Pick $N \in \mathbb{Z}^+$ such that $n \geq N \rightarrow P(|Z_n - Z| \geq \delta) \leq \epsilon$. Let A^+ denote the region bounded by $y \leq z_0, x \geq z_0, y \leq x - \delta$, A^- denote the region bounded by $y \geq z_0, x \leq z_0, y \geq x + \delta$, T^+ the triangle bounded by $y \leq z_0, x \geq z_0, y \geq x - \delta$, T^- the triangle bounded by $y \leq x + \delta, x \leq z_0, y \geq z_0$. Then

$$\begin{aligned}
|F_{Z_n}(z_0) - F_Z(z_0)| &= \left| \int_{A^+} p(x, y) dA - \int_{A^-} p(x, y) dA + \int_{T^+} p(x, y) dA - \int_{T^-} p(x, y) dA \right| \\
&\leq \int_{A^+} p(x, y) dA + \int_{A^-} p(x, y) dA + \int_{T^+} p(x, y) dA + \int_{T^-} p(x, y) dA \\
&= \int_{A^+ \cup A^-} p(x, y) dA + \int_{T^+ \cup T^-} p(x, y) dA \\
&\leq P(|Z_n - Z| \geq \delta) + P(|Z - z_0| \leq \delta) \\
&= P(|Z_n - Z| \geq \delta) + F_Z(z_0 + \delta) - F_Z(z_0 - \delta) \\
&\leq \epsilon + \epsilon = 2\epsilon.
\end{aligned}$$

Thus, $Z_n \xrightarrow{d} Z$.

Linear regression as MLE. Have $p(y|x) = \mathcal{N}(y, \hat{y}(x, w), \sigma^2)$. Then

$$\begin{aligned}
\theta &= \operatorname{argmax} \sum_{i=1}^m -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (y - \hat{y}(x, w))^2 \\
&= \operatorname{argmin} \left(m \log 2\pi\sigma^2 + \frac{1}{\sigma^2} \sum_{i=1}^m (y - \hat{y}(x, w))^2 \right) \\
&= \operatorname{argmin} \left(2m \log \sigma + \frac{1}{\sigma^2} \sum_{i=1}^m (y - \hat{y}(x, w))^2 \right)
\end{aligned}$$

Gradient of cross-entropy loss.

$$\begin{aligned}
\frac{\partial}{\partial z_1} \log \frac{\exp z_1}{\exp z_1 + \exp z_2} &= \frac{\partial}{\partial z_1} (z_1 - \log(\exp z_1 + \exp z_2)) = 1 - \frac{\exp z_1}{\exp z_1 + \exp z_2} = 1 - q_1 \\
\frac{\partial}{\partial z_1} \log \frac{\exp z_2}{\exp z_1 + \exp z_2} &= -\frac{\exp z_1}{\exp z_1 + \exp z_2} = -q_1 \\
\frac{\partial J}{\partial z_1} &= -p_1(1 - q_1) + p_2 q_1 = q_1 - p_1
\end{aligned}$$

In general, let $Z = \sum_i \exp(z_i)$. Then for $i = j$, we have $\frac{\partial q_j}{\partial z_i} = 1 - q_i$. For $i \neq j$, we have $\frac{\partial q_j}{\partial z_i} = -q_i$, so

$$\frac{\partial J}{\partial z_i} = -p_i(1 - q_i) + \sum_{j \neq i} p_j q_i = -p_i(1 - q_i) + (1 - p_i)q_i = q_i - p_i,$$

as desired.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ have Hessian H at x . Then $Hv = \nabla_x((\nabla_x f)^T v)$.

Proof:

$$\begin{aligned}\nabla_x((\nabla_x f)^T v) &= \nabla_x \sum_i (\nabla_x f)_i v_i = \nabla_x \sum_i \frac{\partial f}{\partial x_i} v_i = \sum_i v_i \nabla_x \frac{\partial f}{\partial x_i} \\ &= \sum_i v_i \left(\frac{\partial^2 f}{\partial x_j \partial x_i} \right)_j = Hv.\end{aligned}$$

Suppose we minimize $J(w)$ at w^* . We approximate by a quadratic: $\hat{J}(w) = J(w^*) + \frac{1}{2}(w - w^*)^T H(w - w^*)$. Note that there is no linear term since J is minimal at w^* . We have $\nabla_w \hat{J} = H(w - w^*)$. Including the regularizer, we wish for $\alpha w + H(w - w^*) = 0$, ie $w = (\alpha I + H)^{-1} H w^*$. H is symmetric, so write $H = Q\Lambda Q^T$, so $w = (\alpha I + Q\Lambda Q^T)^{-1} Q\Lambda Q^T w^* = Q(\alpha I + \Lambda)^{-1} \Lambda Q^T w^*$.

Linear regression + regularization. $J(w) = (Xw - y)^T (Xw - y) + \alpha w^T w$. Then

$$\begin{aligned}\nabla J(w) &= 2X^T(Xw - y) + 2\alpha w \\ w &= (X^T X + \alpha I)^{-1} X^T y\end{aligned}$$

L^1 regularization. We have the approximation

$$\begin{aligned}\hat{J}(w; X, y) &= J(w^*; X, y) + \sum_{i=1}^n \frac{1}{2} H_{ii} (w_i - w_i^*)^2 + \alpha |w_i| \\ &\rightarrow H_{ii} (w_i - w_i^*) + \alpha \operatorname{sgn}(w_i) = 0\end{aligned}$$

Suppose $w_i > 0$. Then $H_{ii} (w_i - w_i^*) + \alpha = 0 \rightarrow w_i = (H_{ii} w_i^* - \alpha) / H_{ii}$. If $w_i < 0$, then $w_i = (H_{ii} w_i^* + \alpha) / H_{ii}$. We may also pick $w_i = 0$.

2 RL

Epsilon greedy. Consider k -armed bandit problem, epsilon greedy param ϵ . Suppose estimate stationary is given by p and has unique max at index i . Prob of choosing i is

$$\begin{aligned}(1 - \epsilon)p(i) + \frac{1 - p(i)}{k - 1} &= \left(1 - \epsilon - \frac{1}{k - 1}\right)p(i) + \frac{1}{k - 1} \\ &\leq 1 - \epsilon - \frac{1}{k - 1} + \frac{1}{k - 1} = 1 - \epsilon.\end{aligned}$$

Then long term expectation is

$$\begin{aligned}
(1 - \epsilon)q_*(i) + \frac{\epsilon}{k-1} \sum_{j \neq i} q_*(j) &= (1 - \epsilon)q_*(i) + \frac{\epsilon}{k-1} \left(\sum_{j \in [n]} q_*(j) - q_*(i) \right) \\
&= \left(1 - \epsilon - \frac{\epsilon}{k-1} \right) q_*(i) + \frac{\epsilon}{k-1} \sum_{j \in [n]} q_*(j) = \left(1 - \frac{\epsilon k}{k-1} \right) q_*(i) + \frac{\epsilon}{k-1} \sum_{j \in [n]} q_*(j)
\end{aligned}$$

Suppose instead of $Q(A) \leftarrow Q(A) + \frac{1}{N(A)}(R - Q(A))$, we have the step size α to be constant:

$$\begin{aligned}
Q_{n+1} &= Q_n + \alpha(R_n - Q_n) = \alpha R_n + (1 - \alpha)Q_n \\
&= \alpha R_n + (1 - \alpha)(\alpha R_{n-1} + (1 - \alpha)Q_{n-1}) \\
&= \alpha R_n + (1 - \alpha)\alpha R_{n-1} + (1 - \alpha)^2 Q_{n-1} \\
&= \dots \\
&= \alpha \sum_{i=0}^{n-1} (1 - \alpha)^i R_{n-i} + (1 - \alpha)^n Q_1.
\end{aligned}$$

Wish for the conditions $\sum_{n=1}^{\infty} \alpha_n = \infty$ and $\sum_{n=1}^{\infty} \alpha_n^2 < \infty$.

Gradient-based ascent:

$$\begin{aligned}
H_{t+1}(A_t) &= H_t(A_t) + \alpha(R_t - \bar{R}_t)\pi_t(A_t) \\
H_{t+1}(a) &= H_t(a) - \alpha(R_t - \bar{R}_t)\pi_t(a),
\end{aligned}$$

so $\sum_{a \in \mathcal{A}} H_t(a)$ is invariant of t .

Compute: (fill later).

Fubini's theorem for expectation. Let p, q be indep dist. Then $\mathbb{E}_{x \sim p, y \sim q(\cdot|x)} f(x, y) = \mathbb{E}_{y \sim q, x \sim p(\cdot|y)} f(x, y)$ for a function $f : X \times Y \rightarrow \mathbb{R}$ are equal, and this value will be denoted $\mathbb{E}_{x \sim p, y \sim q} f(x, y)$.

Proof:

$$\begin{aligned}
\mathbb{E}_{x \sim p, y \sim q(\cdot|x)} f(x, y) &= \sum_{x \in X} p(x) \mathbb{E}_{y \sim q(\cdot|x)} [f(x, y)] = \sum_{x \in X} p(x) \sum_{y \in Y} f(x, y) q(y|x) \\
&= \sum_{x \in X} \sum_{y \in Y} f(x, y) p(x) q(y) = \mathbb{E}_{(x, y) \sim (p, q)} f(x, y),
\end{aligned}$$

so this expectation also equals that of f over the joint distribution of p, q .

Let p denote the joint distribution of X, Y, Z . We have $\mathbb{E}[X|Y = y] = \sum_{z \in Z} \mathbb{E}[X|Y = y, Z = z] p(z|y)$.

Proof:

$$\begin{aligned}
\sum_{z \in Z} \mathbb{E}[X|Y = y, Z = z]p(z|y) &= \sum_{z \in Z} \sum_{x \in X} xp(x|y, z)p(z|y) \\
&= \sum_{z \in Z} \sum_{x \in X} x \frac{p(x, y, z)}{p(y, z)} \cdot \frac{p(y, z)}{p(y)} = \sum_{z \in Z} \sum_{x \in X} x \frac{p(x, y, z)}{p(y)} = \sum_{x \in X} xp(x|y) = \mathbb{E}[X|Y = y]
\end{aligned}$$

Now:

$$\begin{aligned}
v_\pi(s) &= \mathbb{E}_\pi[G_t|S_t = s] = \sum_{a \in \mathcal{A}} \Pr_\pi\{A_t = a|S_t = s\} \mathbb{E}_\pi[G_t|S_t = s, A_t = a] \\
&= \sum_{a \in \mathcal{A}} \pi(a|s) q_\pi(s, a)
\end{aligned}$$

And:

$$\begin{aligned}
q_\pi(s, a) &= \mathbb{E}_\pi[G_t|S_t = s, A_t = a] \\
&= \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} \Pr_\pi\{S_{t+1} = s', R_{t+1} = r|S_t = s, A_t = a\} \mathbb{E}_\pi[G_t|S_{t+1} = s', R_{t+1} = r, S_t = s, A_t = a] \\
&= \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r|s, a) (r + \gamma \mathbb{E}_\pi[G_{t+1}|S_{t+1} = s', R_{t+1} = r, S_t = s, A_t = a]) \\
&= \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r|s, a) (r + \gamma \mathbb{E}_\pi[G_{t+1}|S_{t+1} = s']) \\
&= \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r|s, a) (r + \gamma v_\pi(s'))
\end{aligned}$$

Combining:

$$\begin{aligned}
v_\pi(s) &= \sum_{a \in \mathcal{A}} \pi(a|s) q_\pi(s, a) \\
&= \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{r \in \mathcal{R}} \sum_{s' \in \mathcal{S}} p(s', r|s, a) (r + \gamma v_\pi(s'))
\end{aligned}$$

Bellman for action-value:

$$\begin{aligned}
q_\pi(s, a) &= \sum_{r, s'} p(s', r|s, a) (r + \gamma v_\pi(s')) \\
&= \sum_{r, s'} p(s', r|s, a) \left(r + \gamma \sum_{a' \in \mathcal{A}} \pi(a'|s') q_\pi(s', a') \right)
\end{aligned}$$

We define $v_*(s) = \max_\pi v_\pi(s)$ and $q_*(s, a) = \max_\pi q_\pi(s, a)$. We have:

$$v_*(s) = \max_\pi v_\pi(s) = \max_\pi \sum_{a \in \mathcal{A}} \pi(a|s) q_\pi(s, a)$$